Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/15/2022
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2023
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
Contact Email:	gioia.massa@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 11/3/22) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
Task Progress:	The project is currently waiting on the availability of the BRIC-LED hardware and flight opportunities. Fresh seed stocks are being maintained for all the genotypes for the experiment. Meanwhile, progress has been made on analysis of the metadata associated with all the plant spaceflight data deposited at the NASA GeneLab. These data include details about experiment design, hardware, and biological samples and have been curated as a data matrix. The similarities between datasets have then been mapped and the correlations between each set of transcriptomics data further correlated with all the other datasets from the experiments, such as available datasets on patterns of gene expression to reveal patterns of response. This analysis has highlighted assay type (microarray versus RNA-seq) and lighting regime as key variables that need to be controlled for -- to help reveal spaceflight-related patterns in the data. This study has also reinforced the conclusions that oxidative stress and cell wall modifications are key elements of the spaceflight response in plants, showing up in the transcriptomic profiles of multiple spaceflight experiments. Detailed analysis of the publicly available results in GeneLab of experiments performed in the BRIC hardware on Arabidopsis thaliana seedlings has revealed some core response genes (HSP101, ABS4, COR78, PRX39, AAP2 and MAGL4) as being commonly regulated across these spaceflight experiments. Most of these genes map to pathways directly related to oxidative stress response. This BRIC-LED 002 experiment focuses on how mutants in the genes for the proteins Calmodulin-Like 24 (CML24) and the NADPH oxidase RBOHD (an enzyme that generates reactive oxygen species) affect spaceflight responses. These genes are thought to play key roles in pathogen response, but accumulating data now indicates they are also involved in hypoxic and flooding-related responses. As hypoxia is a major predicted effect of spaceflight, research in this project has also continued to analyze how this link between CML24 and RBOHD and response to reduced O2 levels occurs. When inactivated by knockout mutations, both RBOHD and CML24 led to significant resistance to hypoxic stress. These mutants maintained growth under flooding conditions in excess of 4 weeks and suppressed the induction of molecular markers of hypoxic stress, such as upregulation of the gene ALCOHOL DEHYDROGENASE. The current working hypothesis being tested is that CML24 is regulating RBOHD function. This is being approached by assessing the responses of double mutants in both genes to flooding stress and low oxygen environments and to challenge with the plant pathogen Pseudomonas syringae.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Bakshi A, Gilroy S. "Analysis of plant flooding response." Meth. Enzymol. 2022 Nov 11. https://doi.org/10.1016/bs.mie.2022.08.043 , Nov-2022
Articles in Peer-reviewed Journals	Fernandez JC, Gilroy S. "Imaging systemic calcium response and its molecular dissection using virus-induced gene silencing." Meth. Enzymol. 2022 Sep 13. https://doi.org/10.1016/bs.mie.2022.08.006 , Sep-2022

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